BETH ISRAEL DEACONESS MEDICAL CENTER

330 Brookline Avenue, Boston, MA 02215
Internal Medicine - Medical Intensive Care Unit
Tel: (617) 667-7000 | Fax: (617) 667-7100

INPATIENT DISCHARGE SUMMARY - MEDICAL ICU

Patient Name:	Patient cb70e6ae (MIMIC Study ID)
Medical Record #:	MRN-4827561
Date of Birth:	October 8, 2086 (Age: 61 years)
Sex:	Male
Admission Date:	July 15, 2148
Discharge Date:	July 18, 2148
Length of Stay:	3 days
Attending Physician:	Dr. Rebecca Hollingsworth, MD
ICU Team:	Medical ICU Service
Discharge Disposition:	Home with Home Health Services

ADMISSION DIAGNOSIS

Acute hypoxemic respiratory failure secondary to healthcare-associated pneumonia with sepsis.

PRINCIPAL DISCHARGE DIAGNOSES

- Healthcare-associated pneumonia with sepsis, resolved
- Acute hypoxemic respiratory failure, resolved
- Acute kidney injury (AKI) - Stage 2, resolved
- Paroxysmal atrial fibrillation with rapid ventricular response, controlled
- Type 2 diabetes mellitus with polyneuropathy
- Chronic lymphocytic leukemia (CLL), B-cell type, not in remission

SECONDARY DIAGNOSES

- Coronary artery disease, status post coronary artery stenting
- Presence of cardiac pacemaker (dual chamber)
- Essential hypertension
- Hyperlipidemia
- Hypothyroidism
- Obstructive sleep apnea (OSA), CPAP noncompliant
- Morbid obesity, status post bariatric surgery

- History of testicular cancer, treated
- History of transient ischemic attack (TIA)
- Chronic pain syndrome
- Major depressive disorder
- Anxiety disorder
- Glaucoma, unspecified

HOSPITAL COURSE

Brief Summary

The patient is a 61-year-old male with an extensive past medical history including chronic lymphocytic leukemia (CLL) not in remission, coronary artery disease with prior stenting, paroxysmal atrial fibrillation on anticoagulation, type 2 diabetes mellitus on insulin, morbid obesity status post bariatric surgery, obstructive sleep apnea, and multiple other comorbidities who presented to the Emergency Department on July 15, 2148 with a three-day history of progressive dyspnea, productive cough with purulent sputum, fever to 102.3°F, and generalized weakness. The patient had been seen in the outpatient oncology clinic five days prior to admission for routine CLL surveillance and was noted to be stable at that time.

In the Emergency Department, initial vital signs revealed temperature of 101.8°F, heart rate 124 beats per minute (irregularly irregular), blood pressure 92/58 mmHg, respiratory rate 28 breaths per minute, and oxygen saturation 86% on room air. Physical examination was notable for tachypnea, use of accessory respiratory muscles, decreased breath sounds at the right base with crackles extending to the mid-lung field, and irregular heart rhythm. Initial laboratory studies demonstrated leukocytosis (WBC 18,400/µL with 88% neutrophils, 4% bands), acute kidney injury (creatinine 2.1 mg/dL, baseline 1.0-1.2 mg/dL), lactic acidosis (lactate 3.8 mmol/L), and procalcitonin elevated at 4.2 ng/mL. Chest radiograph revealed a right lower lobe infiltrate with possible early effusion. The patient met criteria for sepsis with hypotension and end-organ dysfunction.

The patient was transferred directly to the Medical Intensive Care Unit for management of septic shock secondary to healthcare-associated pneumonia. He was initiated on broad-spectrum antibiotic coverage with vancomycin 1.5 grams intravenously every 12 hours and cefepime 2 grams intravenously every 8 hours after obtaining blood cultures, sputum culture, and urinalysis. Aggressive intravenous fluid resuscitation was commenced with 3 liters of lactated Ringer's solution over the first four hours. Supplemental oxygen was provided via high-flow nasal cannula at 50 liters per minute with FiO2 of 60%, which maintained oxygen saturation above 92%. Serial arterial blood gas measurements showed gradual improvement in oxygenation and correction of metabolic acidosis.

ICU Course by System

Infectious Disease / Pulmonary: Sputum culture obtained on admission grew methicillin-resistant Staphylococcus aureus (MRSA) and Pseudomonas aeruginosa sensitive to cefepime. Blood cultures remained negative. The patient was continued on vancomycin and cefepime with excellent clinical response. Fever defervesced by hospital day 2, and respiratory status improved progressively, allowing transition to nasal cannula oxygen at 4 liters per minute by hospital day 3. Chest radiograph on hospital day 3 demonstrated interval improvement with decreased consolidation. The patient was discharged on a planned 14-day course of antibiotics (vancomycin transitioned to oral linezolid 600 mg twice daily, and intravenous cefepime continued via home health). The Infectious Disease consultation service recommended close outpatient follow-up.

Cardiovascular: The patient developed rapid atrial fibrillation with ventricular rates in the 130s-150s range during the initial septic presentation. Cardiology was consulted. Pacemaker interrogation confirmed normal device function with appropriate sensing and capture. Given hemodynamic stability after fluid resuscitation, rate control was achieved with adjustment of his home metoprolol regimen from 25 mg daily to 50 mg twice daily. He converted to sinus rhythm on hospital day 2. Anticoagulation with warfarin was continued throughout the hospitalization with therapeutic INR maintained between 2.1-2.8. Troponin I was mildly elevated at 0.18 ng/mL on admission but trended down, consistent with demand ischemia rather than acute coronary syndrome. Electrocardiogram showed atrial fibrillation with rapid ventricular response but no acute ST-T wave changes. The patient remained chest pain-free throughout the admission. Blood pressure normalized with fluid resuscitation and remained stable without requirement for vasopressor support.

Renal: Acute kidney injury (creatinine peaked at 2.3 mg/dL on hospital day 1) was attributed to pre-renal azotemia in the setting of sepsis and dehydration. Kidney function improved rapidly with intravenous fluid resuscitation, and creatinine normalized to 1.1 mg/dL by discharge. Urine output was monitored closely and remained adequate throughout the ICU stay. Nephrotoxic medications were avoided, and antibiotic dosing was adjusted for renal function during the period of elevated creatinine.

Endocrine: Type 2 diabetes mellitus was managed with an insulin sliding scale protocol during the acute illness. Blood glucose levels were maintained between 120-180 mg/dL. The patient was transitioned back to his home insulin regimen (insulin glargine 100 units at bedtime and insulin lispro sliding scale with meals) on hospital day 3 with good glycemic control. Thyroid function was stable on levothyroxine 25 mcg daily.

Hematology / Oncology: The patient's chronic lymphocytic leukemia remained stable without acute complications during this admission. Hematology-Oncology was consulted and recommended continuation of outpatient CLL management. Complete blood count on admission showed white blood cell count elevation attributable to infection superimposed on baseline lymphocytosis. Hemoglobin remained stable at 11.2 g/dL (baseline 10.8-11.5 g/dL), and platelet count was adequate at 142,000/µL. Anticoagulation was continued for atrial fibrillation management without bleeding complications.

Pain / Psychiatry: Chronic pain was managed with continuation of home medications including gabapentin 300 mg three times daily. Oxycodone 5 mg was provided as needed for breakthrough pain related to coughing. Psychiatric medications including fluoxetine 20 mg daily, trazodone 100 mg at bedtime, and divalproex 500 mg extended-release twice daily were continued without interruption. The patient's mood and anxiety remained stable throughout the hospitalization.

Consultations

- Infectious Disease - Dr. Martin Chen
- Cardiology - Dr. Elizabeth Kowalski
- Hematology-Oncology - Dr. Jonathan Pierce

DISCHARGE MEDICATIONS

Medication	Dose / Route	Frequency	Duration / Instructions
Linezolid	600 mg PO	Twice daily	Continue for 11 more days (total 14 days); complete course on July 29, 2148
Cefepime	2 grams IV	Every 8 hours	Via home health nursing; continue for 11 more days (total 14 days); complete course on July 29, 2148
Metoprolol tartrate	50 mg PO	Twice daily	INCREASED DOSE for atrial fibrillation management
Warfarin sodium	5 mg PO	Daily at 5 PM	

Medication	Dose / Route	Frequency	Duration / Instructions
			Maintain INR 2.0-3.0; follow up with Coumadin Clinic on July 20
Insulin glargine	100 units SC	Daily at bedtime	Continue home regimen
Insulin lispro	Per sliding scale	With meals and at bedtime	Adjust based on blood glucose monitoring
Aspirin	81 mg PO	Daily	For cardiovascular prophylaxis
Divalproex sodium ER	500 mg PO	Twice daily	For mood stabilization
Fluoxetine	20 mg PO	Daily in morning	For depression
Trazodone	100 mg PO	At bedtime as needed	For insomnia
Gabapentin	300 mg PO	Three times daily	For neuropathic pain
Oxycodone	5 mg PO	Every 6 hours as needed	For moderate pain; use caution, avoid driving
Acetaminophen	325 mg PO	Every 6 hours as needed	For mild pain or fever; maximum 3000 mg per 24 hours
Levothyroxine	25 mcg PO	Daily in morning	Take on empty stomach, 30 minutes before breakfast
Pantoprazole	40 mg PO	Daily	For gastric protection
Docusate sodium	100 mg PO	Twice daily	Stool softener
Senna	8.6 mg (2 tablets) PO	At bedtime as needed	For constipation
Ziprasidone	40 mg PO	At bedtime	For anxiety; take with food
Latanoprost eye drops	0.005% (1 drop each eye)	At bedtime	For glaucoma management

DISCONTINUED MEDICATIONS

- Metoprolol 25 mg daily - DISCONTINUED; replaced with metoprolol 50 mg twice daily for improved rate control
- Hydrocodone-acetaminophen 5-325 mg - DISCONTINUED; replaced with separate oxycodone and acetaminophen for better pain control titration

DISCHARGE INSTRUCTIONS

Activity

Activity as tolerated. Gradual return to baseline activity level. Avoid strenuous exercise for two weeks. Continue ambulatory activities within the home. Use of CPAP machine is strongly encouraged during sleep.

Diet

Diabetic diet, cardiac heart-healthy diet (low sodium, low saturated fat). Monitor carbohydrate intake. Maintain adequate hydration with water intake of at least 1.5-2 liters per day unless otherwise restricted.

Wound Care / Medical Equipment

PICC line (peripherally inserted central catheter) placed in right upper arm on July 16, 2148, for home intravenous antibiotic therapy. Home health nursing will provide PICC line care, dressing changes every 7 days or as needed, and intravenous cefepime administration. PICC line to be removed after completion of antibiotic course on approximately July 29, 2148.

Self-Monitoring

- Monitor blood glucose levels four times daily (fasting, before meals, and at bedtime) and record in logbook
- Monitor temperature twice daily; call physician if temperature exceeds 100.4°F
- Monitor for signs of worsening infection: increased cough, shortness of breath, chest pain, or confusion
- Weigh daily and record; report weight gain of more than 3 pounds in one day or 5 pounds in one week
- INR monitoring via home health or Coumadin Clinic as scheduled

Warning Signs - Call 911 or Go to Emergency Department if:

- Severe shortness of breath or difficulty breathing
- Chest pain or pressure
- Severe bleeding or blood in stool or urine
- Signs of stroke: facial drooping, arm weakness, speech difficulty
- Altered mental status or loss of consciousness
- High fever (greater than 103°F) not responding to acetaminophen
- Severe pain not controlled with prescribed medications

Call Primary Care Physician or On-Call Provider if:

- Temperature above 100.4°F
- Increased cough, shortness of breath, or chest discomfort
- Redness, swelling, drainage, or pain at PICC line site
- Nausea, vomiting, or inability to take medications
- Blood glucose consistently above 250 mg/dL or below 70 mg/dL
- Signs of bleeding: easy bruising, bleeding gums, blood in urine or stool
- Any other concerns about recovery

PENDING STUDIES AND RESULTS

- Final antibiotic susceptibility testing on sputum culture - results pending as of discharge; will be followed by Infectious Disease team
- Repeat chest radiograph scheduled for July 22, 2148 (outpatient) to document resolution of pneumonia
- Pacemaker remote monitoring report due July 25, 2148 (Cardiology will review)

FOLLOW-UP APPOINTMENTS

Provider / Clinic	Date / Time	Location	Purpose
Coumadin Clinic	July 20, 2148, 9:00 AM	BIDMC - Anticoagulation Clinic, 4th Floor	INR check and warfarin dose adjustment
Dr. Samuel Rothstein (PCP)	July 23, 2148, 2:30 PM	BIDMC - Internal Medicine Associates	Post-discharge follow-up, medication review
Dr. Martin Chen (Infectious Disease)	July 26, 2148, 10:00 AM	BIDMC - Infectious Disease Clinic, 5th Floor	Review antibiotic response, plan for completion of therapy
Dr. Jonathan Pierce (Hematology-Oncology)	August 2, 2148, 1:00 PM	BIDMC - Cancer Center	Routine CLL surveillance, assessment after acute illness
Dr. Elizabeth Kowalski (Cardiology)	August 9, 2148, 11:00 AM	BIDMC - Cardiology Clinic, 3rd	